

Ice Breaker

Do you know these?



VS

Do you know this?



cvc-shared-tools

Climate adaptation vs. Code adaptation

"We model climate adaptation, but how does our modeling code adapt?"

Reference: *Agentic Workflows for Climate Modeling* (arXiv:2606.25076)

3. Why `cvc-shared-tools` exists

The Problem

- **No discoverability:** Useful scripts are scattered in personal `/gpfs/scratch/` dirs.
- **Copy-paste culture:** Workflows branch and diverge.
- **Lost knowledge:** When authors leave, scripts are abandoned.
- **Brittle parameters:** Hardcoded absolute paths and modules break after HPC updates.

The Solution (`cvc-shared-tools`)

A centralized, curated repository for:

- Pre-processing tools
- Interpolation workflows
- Common Earth System scripts

Goal: Almost-zero-cost maintenance through standard practices and automated documentation.

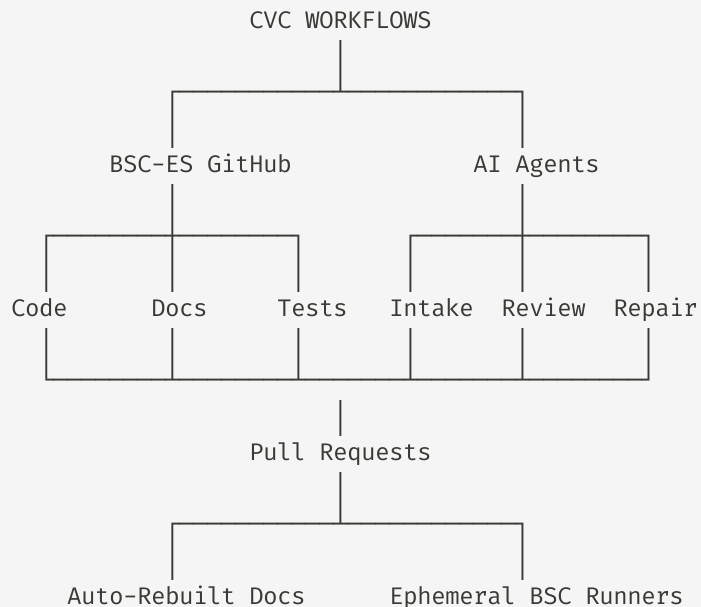
4. Standardizing the workflow contract

To fix the maintenance burden, every workflow must have a machine-readable contract: `metadata.yaml`.

```
schema_version: "1.0"
workflow:
  name: "en4-to-orca"
  description: "Interpolates EN4 ocean data to ORCA grids"
environment:
  target_platform: "mn5"
modules:
  - "netcdf/4.7"
  - "cdo"
  - "sosie"
execution:
  script: "scripts/run.sh"
```

This `metadata.yaml` drives everything: CI checks, index generation, and agent ingestion.

5. The Workflow Architecture



An ecosystem where AI proposes structure and humans validate science.

6. Free, Auto-Rebuilding Documentation

Zero Manual Copy-Paste

- Scientist pushes `metadata.yaml` updates.
- CI script (`update_workflows_index.py`) runs on every commit.
- Automatically generates `WORKFLOWS.md` .

The Resulting Portal

- **Searchable:** Filter by `MN5` , `CD0` , `NEMO` .
- **Status Badges:** See validation state instantly.
- Rebuilt in 3 seconds.
- Hosted on ReadTheDocs for free.

7. Trust has multiple dimensions

A workflow can be:

Dimension	State
Documentation	reviewed
Software checks	passed
Scientific validation	required
Reproducibility	partial
HPC environment	unknown
Maintenance	current

Do not collapse all of this into one `status: verified`.

8. AI is not the source of truth

The agent should create:

9. What an ingestion interaction looks like

A scientist submits a GitHub issue:

"Please ingest workflows/prediction-data/foo for reuse."

The Pipeline:

1. gh-aw + LLM (Claude/Gemini)
2. inspect repository & extract evidence
3. generate metadata & docs
4. run deterministic checks
5. draft pull request
6. human review

The Interface: The scientist does **not** need to learn an AI framework.

GitHub is the interface.

10. The user can choose any local AI agent

Climate Commons should not own a local AI runtime.

Institutional automation

= GitHub + gh-aw + Gemini

Personal productivity

= scientist's preferred agent

agy / Antigravity

Copilot

Claude Code

The repository should not care which local agent produced a Git diff. It supplies the contract (`AGENTS.md` , schemas, CI checks).

11. Security & Safety Boundaries

Agents may 

- inspect code and docs
- generate metadata
- propose documentation
- detect drift & propose fixes
- create draft PRs

Agents may not 

- merge to `main` autonomously
- launch production HPC jobs
- change scientific methodology
- modify production infrastructure
- access unrestricted secrets

The intended control loop is: Read → Reason → Propose → Check → Review → Merge

12. What the prototype will actually test

First vertical slice

Workflow ingestion + reuse documentation (Start with ~10 workflows).

Success is not "the README looks good."

Success is:

Can another scientist use the workflow without asking its original author the basic questions?

13. What comes after the first vertical slice?

C — Maintenance

Use scheduled agents to detect documentation drift, broken links, stale dependencies, and orphaned workflows.

D — Infrastructure migration

Identify workflows affected by HPC changes, compare environment assumptions, and prepare migration PRs.

Do not build C or D until ingestion demonstrates value.

14. Why this can stay low-maintenance

The system deliberately avoids becoming a platform to operate.

No

- custom web application
- always-on agent service
- vector database / RAG backend
- mandatory local CLI
- autonomous HPC service

Yes

- GitHub & GitHub Actions
- `gh-aw`
- Repository metadata
- Deterministic CI
- Pull requests

Git remains the durable artifact. Agent vendors remain replaceable.

15. The first implementation steps

Phase 0

Repository Trust

- • Protect main branch
- • Pin Actions
- • Establish security files

Phase 1

Deterministic Base

- • Workflow schemas
- • Auto-index scripts
- • CI checks setup

Phase 2

Agentic Prototype

- • Install gh-aw
- • Read-only health audits
- • Ingestion → Draft PR

16. Decision criteria

Does it save time? • Does it improve reuse? • Does it reduce knowledge debt?

"The project succeeds when the repository becomes more useful than the sum of the scripts inside it."

Join the effort

`https://gitlab.earth.bsc.es/cvc/cvc-shared-tools.git`

Thank You

Questions & Discussion